Julius Plücker — *Neue Geometrie des Raumes* (1868–1869)

The Boneyard is a collection of works left unfinished, unpublished, or unread at their authors' deaths — dry bones in the valley, in Ezekiel's image. Each presentation tells the scholar's story, weighs what survives of his work, and states plainly what completing it would take, so that a book the world never received whole can be met at last. Each of them was one of one. There will never be another like him, and his light will not shine again except through the work he left behind.

A Boneyard dossier — researched and presented by Robert Tilford, Hays, Kansas. Filed 14 October 2026.

Disclosure: prepared with AI assistance — the six subject reviewers named in the index are AI agents, each desk-checked at this desk; all rulings and final decisions are the author's own.

Dedication

To Julius Plücker (1801–1868), who lost nearly twenty years of mathematical work and found his way back near the end of his life. A feud with Steiner, who declared he would stop writing for Crelle's Journal if Plücker continued in it, helped push him out of mathematics and into physics. When he returned to geometry, he built a new geometry of space on the straight line itself as an element of space. He died on 22 May 1868 with the work unfinished, and his former assistant Felix Klein completed and published the second part the following year. This edition is dedicated to Plücker, with Klein's completing hand credited where it belongs.

The story

Julius Plücker (1801–1868) lost nearly twenty years of mathematical work to a feud, returned to geometry only near the end of his life, and then died before his last book was finished. That sequence is also why the book exists in the form it does: part Plücker, part the work of the younger man who completed it for him.

The feud was with Steiner. Steiner declared he would stop writing for Crelle's *Journal* if Plücker continued in it. That declaration helped push Plücker out of

mathematics and into physics for nearly twenty years. The record says it helped to push him out; it does not present the feud as the sole cause, and neither does this edition.

When Plücker returned to geometry, he returned to it late. His last work set out a new geometry of space, founded on considering the straight line itself as an element of space, rather than building space only from points. In that work, the theory of complexes of the second degree was left unfinished by his death, on 22 May 1868, and was continued by Felix Klein.

Klein was Plücker's former assistant. He edited the second part and published it the following year. So the book that bears Plücker's name is, in its second volume, a collaboration completed after its author's death, and any honest English edition has to show where each man's work stands.

The work and what survives

The full title is *Neue Geometrie des Raumes gegründet auf die Betrachtung der geraden Linie als Raumelement* — a new geometry of space, founded on the consideration of the straight line as an element of space.

The Erste Abteilung, the first part, was published by Teubner in 1868. The Zweite Abteilung, the second part, edited by F. Klein, was published by Teubner in 1869. Plücker died before completing the second volume; Klein, his former assistant, edited and published it the following year.

A scan of the 1868 Erste Abteilung is on HathiTrust, with the identifier `wu.89062907357`, cited in a Springer *Archive for History of Exact Sciences* reference list. The German text of 1868–69 is in the public domain.

Its English fortunes

No English translation of the book found after research.

A boundary must be stated plainly, because it limits that finding exactly. Plücker's founding 1865 memoir, "On a New Geometry of Space," was published in English in the *Philosophical Transactions*, 155 (1865): 725–791. The seed memoir is already English; the two-volume book built on it is not.

An English reader can therefore read the announcement of the new geometry, so to speak, but not the full work in which Plücker, and then Klein, developed it.

What completing it would take

The completable form is to translate both Abteilungen as one volume. The work is of medium length, single-source, and self-contained.

One condition governs the whole undertaking. Klein's editorial completion must be credited to Klein section by section. He is a co-maker of vol. 2, not a silent hand. That matters for fairness as well as for scholarship: a reader needs to know which parts Plücker left, which parts Klein completed, and on what basis the completed text stands. Crediting each author for the work that is his is the rule of this collection, and this book is a clear case for it.

Sources

- https://en.wikisource.org/wiki/A_History_of_Mathematics_(1893)/Recent_Times/Analytic_Geometry (the Steiner feud; Klein continuation)
- https://arxiv.org/pdf/1912.07138v1.pdf (edition history; 1865 memoir reference)
- https://link.springer.com/article/10.1007/s00407-023-00305-1 (HathiTrust ID citation)